

ActInf GuestStream 068.1 ~ Darius Parvizi-Wayne, "Distrusting the policy"

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hello and welcome this is otive inference guest stream 68.1 it's January 23rd 2024 and we're live with Darius parvis Wayne it's not a podcast I am confused but it will be fun yes thank you Darius for coming on feel free to uh introduce your and the work looking forward to the presentation and discussion thank you Daniel yeah it's a bit it's a bit odd of an experience to be on this side of the camera and also for it to be live because I'm used to having the benefits of uh of sort of post talk editing although I don't do very much of that um so I'm Dar P way and I finished my masters at UCL last September um I'm currently working at Royal hallay in a cognitive Neuroscience lab in my own private work I work in active inference you may have seen me on the active inference insights podcast that I do with the Institute and I've recently written this paper um called distrusting the policy how inference over action shapes our experience of temporality in flow States and life more broadly this is kind of a followup from a paper that was submitted earlier last year with l s Smith Ry pittia yakob lowski miles Toof and Carl friston um and this is a follow-up piece to that before I start I'm aware that it's a Punchy title uh and I just want to clarify that I am not proposing to explain temporality or the experience of time entirely because it's an extremely rich um area of study in which physics and philosophy and cognitive science converge and I do not have the time the effort or the energy to really tackle all of those problems so it's going to be rather specific but I'll make clear where I'm purposefully missing stuff out okay excellent so there's kind of a central question before we even get going which is what is time and I want to clarify that I'm not talking really about so-called objective time now I know that's itself a tendentious and tricky claim to make because of Relativity but we can have a kind of simple definition given to us actually by some wonderful researchers um botai Jabara who have also worked on active inference accounts of subjective temporality they Define objective time as the temporally sequential development of worldly processes bound relatively to space and so that will incorporate space time as a as a concept in physics so it's spatial in the sense that it follows

spatial laws and it's indivisible and then it's directional now this is a little bit more controversial uh but this ties time to thermodynamic entropy which goes up in close stays the same or goes up in close systems second law of Thermodynamics and this is the So-Cal thermal time hypothesis uh Carelli Lee smolen those are the sort of researchers to go to I as I said at the beginning there are obviously diverse accounts of time especially in phys uh philosophy of time and the physics of time so three main contenders are presentism um which states that only the present exists the growing block Universe which states that the past and the present exist and the future is yet to exist and the internalism which says that basically they're all happening now so din uh you know I believe that there are dinosaurs is still true um I don't really want to delve into those if people do um I would start with the work of someone like John mtag who was very instrumental in developing the philosophy of time and of course there's a lot bit work being done by K relli uh among others on the physics of time but what I'm more interested in today is subjective time time is experienced time is a phenomenon in the sense of being phenomenological now it seemed to me when I started writing this paper that everyone had made the assumption that it flows that there's a what's called a passage phenomenology um and I and my account is predicated on the idea that time flows that said there is a question mark after that because turns out there's a fair uh dose of skepticism about this notion that time flows even within phenomenal States so this is called well this could be ConEd as uh skepticism over passage phenomenology rather than objective passage and these are people like calendar Deng and Miller um so for example Miller thinks actually it's our linguistic structures which give us that impression Deng thinks it it's just successive temporal events without this notion of flow or passage um this paper and this talk I guess can act as a repost to that again I'm not going to delve too much into their arguments uh I would just like to take it at least for this talk that needs to be treated as some sort of axiom that it's flowing and then the other thing is that it's thick and that's a slightly strange adjective um for me to use here especially for people who aren't familiar with the cognitive science of time what does it mean for temporality to be thick well that leads us very nicely onto this man this is Edmund hustle who was the sort of The Godfather the progenitor of phenomenology he taught haiger and so the lineage really starts with him and I also have to mention William James because he's very important in this and William James actually he came up with the notion of what's called a specious present and what we could call a thick present and this is a notion that well we when we divorce objective time and subjective time we can cast objective time as following this kind of linear sequential uh order so going back to the botai and Jabara

definition we have a then B then C now hle in his examinations of subjective experience and how and phenomenality realized or recognized or postulated that the present moment is not just this uh in this divisible uh unit of time that can be separated from the past and the present rather what makes up the present moment is itself a constitution of what he call the Primal impression which we can argue is this or we're going to talk about that in a sec this very narrow slice of nness but also what has just passed and what is about to become so for example this is a diagram is an adaptation from one of H's diagrams because heel is also a mathematician but as we can see here let's take this time point B so this is objective time point B well H we're saying here is that b is also constituted by what's called a retention sorry uh yes a retention of what's just happened soor we're looking at this line here so a retention of what just happened this is a so a retention of what's happened a but see how here it's in Brackets it's a part of b and a sort of what equals a protention what we could say an active influence is a prediction of what's about to happen and so this would be C now C in terms of objective time hasn't actually happened yet but the sort of expectations which are to be fulfilled or not fulfilled um constitute the experience of B and so do the uh residual memory in some sense of a now hus makes it very clear that this this uh account of temporality is very different from uh explicit recollection or expectation or imagining rather this this uh structure is the very structure that defines time Consciousness and so we can kind of see in some sense how flow already comes out of this in so far as protentions are fulfilled or unfulfilled and therefore become retentions which are which recede back in time but also have this predictive element i. they REM i. they are prens what's going to happen and I think to get a little bit more clarity on this picture it's worth just discussing this notion of a primal impression so I think the Primal impression is for my reading and I highly recommend this gallager a harvey paper because I think it's incredibly Illuminating the Primal impression most people take as real and as constituting what gagans or Harvey called the component of Consciousness that is narrowly directed towards the now phase of the object okay so it's a little bit vague what that means and actually I think it's more parsimonious and slightly more elegant to say that it's an abstraction uh and this is actually kind of in line with der's uh reading of hle where he distinguished the theoretical construct of the primary impression which is useful for us in modeling and in an instrumental way from the actual thing that appears in phenomenon the phological Pres um world is always horrible to say and so my argument here in some way is that the nowness what now is is in a sense already anticipated it's it's it's already anticipated because it's being proted right and that protention is is in a sense a retention already because it's projecting into

the next step and it's already predictive for what's going to come at $t+1$ so we can treat the present moment itself as the result of prediction and then the fact prediction about the next step and the fact that that prediction is going to be uh fulfilled or unfulfilled now this before I get on to the flow of time uh I recognize that this is complicated and it requires uh some nonlinear ways of thinking I highly recommend bogar anj's 2023 paper uh where they actually give a very nice base graph and active inference account of this exact way of combining objective time and subjective time in a hylan way uh and I think that should clarify exactly what I mean here again I'm not actually too interested in deriving a perfect structure in an active inference sense a perfect model of hles time I think that's been done very well by botar and Jabara and also people like ma Alber rassin and her team what I'm more interested is in the flow of time per se and what's the difference there well this is the structure of the present as is as time Consciousness as is I'm more interested in the fact that time seems to flow that the present seems to flow into the past and the future seems to flow into the present and that itself is a bit more of a dionic process so that takes me to the account of time's flow now whereas the structure of time Consciousness as I said has been explored quite comprehensively uh within the active inference community and Beyond the notion of time's flow perhaps because some are skeptical that time does flow has itself Maybe being touched upon a little bit less the most famous or the most um yeah I guess the most concrete proposal comes from yakob Hoy and his colleagues and they argue that the flow of time arises as a system gives up one hypothesis and settles on a new one due to its propensity for distrusting the present so this is a rather famous 2015 paper called distrusting the present and the notion here is that every hypothesis every state hypothesis I have about my observation is going to be really good for that observation but immediately it's going to become less and less good because the world changes and so because I have a higher order expectation that the world is going to change it's actually adaptive it's in my best interest as a free energy minimizing system to start to distrust this hypothesis because it's not going to be capable of explaining the sensations the observations at $t+1$ so this is put very nicely by them as the hierarchical system settles on this hypothesis which for them is best at minimizing prediction error this is continuous State space stuff so that's totally fine it needs to begin looking for a new and better hypothesis and again that's contingent on this higher order expectation that the world will change crucially uh and we will I will explain this in more depth the mechanism that I think yakob is proposing here is is perceptual and attentional uh and we would cast that as the a matrix um within pomdp schemes in active inference and that really casts or encodes the uh

probability distribution of the likelihood so that is what is the likelihood of um this state given my observations No Way Around what is the likelihood of this observation given my state the posterior is x given Y and the idea here is that a um the observation is uh at time T is yielding precise precise um precise likelihoods that are going to govern a hypothesis but because the attentional spotlight is always moving around and the world is changing that observation itself is going to lose precision and so the new observations which yield different hypothesis are going to gain precision and therefore the system needs to update but I think most crucially is this notion that it's perceptual and it's about state or hypo as he calls it a hypothesis updating a state hypothesis updating and crucially for us he also yakob and his colleagues mention also the flow uh the speed of the flow of time so we're now adding another dimension to this phenomenology um and I won't read out the whole thing because that's a bit dull but basically he's saying that the faster or the more the more frequently one up uh one updates one's hypothesis about the world the faster time seems to flow so for example we he gives the example of a war zone well in a war zone and I think this is totally reasonable the perceptual hypothesizing has to keep updating because it's volatile so one moment I might be perceiving X and the next moment I might be perceiving y but that's that the frequency of that is happening a lot faster in a different scenario such as waiting an adult airport the idea would be that I don't have that I don't expect the world to be that volatile because maybe I'm just staring at a wall and so I don't have to do this perceptual updating or this hypothesis updating um as ycob describes it now what's the problem uh and this is n years old and I haven't really seen anyone say that there's a problem um but I thought there might be a problem and I problem in the sense is the fact that this account is fundamentally perceptual and so there are three T three situations I want to bring up here and then I'll talk about how they pertain um to the question at hand one is what I call St uh static predictability so this would be something like staring at a wall for a prolong period of time so that's our middle picture here agree that time would pass rather slowly there now the second one is so-called Dynamic predictability now these are my terms I'm very happy for be able to critique them but I'm calling this Dynamic predictability now this would be the chaps on the right these are those people who may have played the video game level and enjoyed it the first time but for some reason uh some torturous devil is making them play over that video game level over and over and over again okay so that's Dynamic predictability I'm not going to talk about how it pertains right now to the flow of time and then the other one is flow uh flow States and again I would Point people in the direction of the paper that I wrote with my with my great colleagues last year on Flow States

but this is slightly different in the sense that the world the volatility of the world is inferred there believed to be a sufficient level of volatility such that I have these uh narrow action policies these temporally short action policies and this adaptive of these fast flows of um motor responses to the world so it's um so that's flow stat and you'll get that canonically in um context such as surfing rock climbing um and so on so this is the work of 6s Mahal and others John thei and hopefully you know we've contributed something as well now um let's just very quickly touch on the speed of time and flow um people uh this is a ni study by rre and colleagues in 2021 and they found that the more flow of participants experience the faster time passed objectively however flow did not correlate with duration estimates now that might seem like kind of the same thing so how are they dissociable well I think the easiest way to think about this is what questions would you ask to discover these facts you would ask for the first one did time how you know how quickly did time pass subjectively for you and then the second question would be how long do you think that took and interestingly in flow we get this sense that time is speeding up but we don't actually uh we actually have quite accurate estimations of duration so for me this shows that these phenomenon phenomena are dissociable and motivates my focus on the subjective speed of time and flow because there something going on um this is this is not particularly incontrovertible um I don't think because sixense Mahal it it goes right back to his qualitative work uh the notion of time dilation has been present in the literature for a while but this is actually one of the first concrete um pieces of experimental work that has validated that and showed that inflow time seems to pass faster okay now I want to go back to this not this woman um in static predictability staring at the wall the there are some nice studies here which basically show that time does seem to slow down in these context um so fine we both agree on that now what's the problem well remember the the model of distrusting the present is fundamentally perceptual um and so you can see here uh yakob and his colleagues uh account of distrusting the present in terms of perceptual attention um and this notion of attentional spotlighting again um it'll be much easier for you to read it than for me to explain it but I think the critical point here is to take it back to here according to yob these video game boys they're going to be in context where they're getting still volatile perceptual uh um information in the sense that I might be playing a video game where things are changing the whole time but I've still played it a 100 times so I'm bored and I would postulate that I think time is going to move pretty slowly there and in flow States we've seen that time seems to move rather quickly but according to this purely perceptual account there should be no strong difference between the sense of time and flow and these

dynamically predictable situations because because in both of them you could find a situation where there's the same or even more State updating to do in dynamically predictable situations I think an even stronger example of this I actually put uh in the paper so there is a paper on this um where my thoughts are going to be a little bit more Lucid because obviously this this is not scripted um and the example I give there is of the poor person who has to stare at a screen with a DOT that constantly changing color now there's a lot of perceptual State updating there and let's say it's randomly changing color so we can't have an action policy that can pick out the patterns and say um you know you can have these extended action policies that say okay it's going to go red green yellow red green yellow so it's constantly changing color now I I was hard pressed there was no empirical work that contrasted flow and dynamically predictable situations uh and so this is empirical work that I would like to do and I'll get on to that but I would still propose just uh heris and intuitively that time would seem to slow uh flow rather slowly I think another issue with this is how we if we go back to his original example of the war zone and the do airport even in the context of St uh static predictability there's always low-level State updating that's going on right the world is never static so even if it's the fact that uh the paint on the wall is very slowly cracking or there's a fly that passes in front of you like the the actual uh the world itself never changes and howwi doesn't really give an account of exactly why or how that can be incorporated into his model so this made me think well I don't think the perceptual hypothesis is necessarily the most elegant or fits the phenomenology perfectly and I think it's incred I'm not I I love yakob um I think it's an incredible piece of work but I just think there's a little bit more that could be added so this is where I came in with this notion of distrusting the policy okay so this is a base graph um this is actually from L SED Smith's uh 2021 work uh I won't go too into it although it's actually not too difficult but basically what what what is shown here is the fact that in Act of inference and the references are there we talk about action policy selection so policies are these sequences of action we talk about these in terms of tree searches so I might have multiple potential policies that I could pursue and I basically calculate expected free energy uh which we can have as um epistemic gain and pragmatic gain um if you want I can sort of talk a little bit more about that but I don't think we need to get into the the theory too much I talk about we have these Tre searches where the agent evaluates the possible consequences of these different policies in terms of expected free energy now they're not just doing this on a whim as you can see here at the top of the graph that lar gives us L and his colleagues because it was a collaborative paper um the expected free energy is uh constituted in part by or or decided upon

in part calculated by the So-Cal C Matrix the C Matrix are the preferences about the sensory outcomes that you have so if I have very highly precise sensory expectations about how my action is going to the sensory outcomes of my action that's going to drive μ and it's in you know and it's in favor with my generative model that's going to drive that pragmatic action μ indeed the pragmatic aspect of the expect free energy formula is the probability of why the observation given c μ there is also this e Matrix which is habits μ we can take as habits prior as over policies μ and we'll see actually that that comes in a little bit more in flow but I think the main idea here is well there's a couple things one is that an action policy need not just yield need not just be isomorphic or map onto just one observation as you can see here we can have action policies that stretch deep into the future as long as we as long as we presume that the future is predictable such that μ we have we have an N plus one of the so-called B Matrix the B Matrix are State transitions so I might have an action policy where I go okay I'm going to be in this state then this state then this state and obviously this is informed in real time by the observations I make uh through the a matrix through the likelihood Matrix but the point here is is that μ the tree search can like a tree be extended or narrow okay so that's one thing the other thing to not here is that every time I make an action policy or every time I decide upon an action policy these things like the E Matrix and the C Matrix are implicated so we do not just blindly select an action μ this goes right back to the work of Chris Fieff and his colleagues at UCL in the late 90s on uh actually how action happens and forward modeling but the idea is that beliefs about the consequence of our action are fundamental to even doing action in the first place okay so that's that's the idea here in terms of the actual theoretical side of it now what does that mean in terms of the flow of time so coming back to the flow of time so recall that in flow I mentioned it rather briefly but the idea is that the world is volatile and although we have a good grasp of it although we're occupying or embodying uh what Kistler and others and Brunberg and right Val called a optimal grip we still need to be aware that the world can change very quickly so I might slightly have my bow angle wrong if I'm playing the violin or the wave might not turn exactly how I expect it to and what that means is a deep temporal planning with these deep trees is precluded in flow now what that means is if we take five minutes of flow and we can we can assume that flow does last that long and there are some nice empirical studies on that per objective time unit I'm going to have to be doing more bouts of planning than I would let's say in so-called μ static static predictable context and sorry μ and deeply predictable uh and dynamically predictable contexts so in that case what what we end up reworking is that the flow of time is not about distrusting the present

hypothesis about the sensory data per se because in dynamically predictable context and in flow states that might be the same but rather the iterative speed of belief updating about the current action policy now a question that kind of came to mind mind okay that's that sounds okay but we've been talking about perception the whole time and arguably if we're going to stick to the retention primary impression and protention scheme we're going to have to talk about perception uh and indeed you can see gallago and Harvey's um sort of uh Point here um that actually hustle analysis is very much characterizing perception and obviously the perception of time itself is a perception right we're talking about the phenomenology of time the phenomenology of Passage time that's a perceptual act so how are we going to ground uh planning and policy selection how we going to fold that into this picture well it comes right back to what I said before and it's the fact that policy selection includes the deployment or the uh Precision of certain beliefs in a generative model including predictions over preferred Sensations which is uh encoded in the C Matrix and state transitions which are encoded in the B Matrix so if you look on the right side of the page here this is actually a base graph from my paper something I did with Lars and you'll see here that if um so it's rather complicated but actually I'll go through the text first and then we'll come back to the base graph um again the idea is actually at least in flow so maybe it's actually worth looking at it um what we what flow really entails is a contextual cue for mental action so what you see here it's um it's in the mental state inference box it's you with the two at the top and the one that's an action that's triggered by E2 at the top uh with the superscript and that's a habit but that's not that's a habit of driving action but not a physical action per se what that mental action do uh is doing is driving aood distribution the a matrix and so again what that means really for this for this example or or for the broader paper um is that mental action is not limited just to the selection of policies but also these Precision way of beliefs about sensory out comes so what does that mean well it means that if I do iterative uh policy updating if I do more iterative policy updating per objective unit of time then what I'm also doing is I'm doing more updating about my preferred sensory uh expectations I.E my cmatrix is getting updated more frequently and that's really the critical point here um so for example in flow what you're getting is you're getting this constant uh you can see here that the The e- Matrix and the the C Matrix fold into the action policy because this uh planning would be taken as to be rather shallow you're going to have this process iterate more frequently over a given period of time now just to clarify for people uh who might not have any idea what's going on in this Bas graph as I said the idea here is that um the mental action is deploying um it's about deploying Precision over these uh

Precision beliefs so you have um the likelihood mapping the Precision of the transition mapping and the Precision of preferences note that basically you can't the idea here is that you can't necessarily perceive Precision so the way that that's getting fed back up to the higher level is through the observations made at 0 subscript to superscript one um so that's actually stated in the italics um but I think yeah the the point here I'm very happy to go into this this is sort of deep parametric modeling which has um been going on with L's work and Casper's work Casper hesp and I think it's wonderful and I'd love to talk more about it in terms of this though I think it's a very simple point that these C matrices whether it's C with the two as a superscript or see with one as the superscript which is really going to be the kind of critical see and flow as the one with the one superscript because that's the one which basically drives the um sorry no no I'm talking about the precise beliefs of the sensory expectations that's the one that's going to drive the action policy the the U so to speak um that's just going to get updated more frequently in flow States because of the shallow action policies that flow States in gender so therefore what's the kind of ultimate uh conclusion here well my conclusion was that underline the speed of temporal temporal flows since beliefs in coding the C Matrix now this is kind of the critical point are fundamentally predictive I.E they preend um and um and they are also retained over a period of time why is that it's because my sensory expectations are going to govern what I perceive as we know from active inference because the uh posterior really is just this negotiation of a prior expectation and the likelihood distribution and yet they are also informed by just past experiences are. they involve a form of retention because as you saw with the flow that c Matrix the sensory uh the sensory exp about the consequences of my action are fundamentally Downstream on this contextual cue so I think there's a nice way here in which we can wrap sensory expectations in coding the C Matrix in the jelen phenomenology again the really critical point here is that this is this can be independent of the pure perceptual inferences and pureal inferences here are kind of the combining about prior beliefs about States um before any observations the So-Cal D Matrix and a Precision weighted likelihood distribution the a matrix so that's happening at the very top here of the contextual state inference in these um dynamically predictable uh situations and um in flow States we might have the same number of pure perceptual inferences but I would posit that in flow time seems to move faster and predictable them predictably Dynamic situations and the reason I think that is is because we should be talking about perceptual uh inference updating about sens expectations with respect to the C Matrix uh and not perceptual inference about the combining of D Matrix and the a matrix um and again this is because that uh the frequency with which

we update those beliefs encode in the C Matrix is contingent on the frequency of policy planning so just to conclude um in terms of going forward uh this is again so just to make it very clear that it's a pre-print right now this paper I I'm not claiming that this is the comprehensive account of uh temporality or the speed of the flow of time but it's a suggestion so so um I'd love to get feedback from people but in terms of going forward I think this would be an interesting thing to test whether we could set up a situation where um we have one group of you know what your three groups you have a controlled group who aren't doing much um a second group who are playing a video game for the hundredth time it's predictably uh dynamically predictable and then flow States uh which can also be done through something like a video game and what I would expect to see just at a very basic level is that um in flow we could replicate that finding of time moving faster subjectively and confirm what I think is a reasonable intuition that in these dynamically predictable States time seems to move slowly or at least slower relative to flow States okay that's me Daniel feel free to far away excellent all right great times it flew by I was paying attention yeah well well it's very well another part of this I mean I missed quite a lot out that's part of the paper the paper's relatively large is that attention itself is an action uh and this is often forgotten by people but attention is a is an action it's a mental action and that itself will have its Associated C Matrix so even paying attention itself would constitute a mental action one one note on that and then um anyone who has a question live chat I'll read um policy selection under expected free energy is shaped by pragmatic value aligning future observations with your preferences and epistemic value Information Gain so that's very interesting that attention as internal action or covert action could be driven also by pragmatic attentive considerations here's the kind of regime of attention I expect to be in and also there could be epistemic or exploratory regimes of attention that maybe like reflect us getting distracted just to gain information even when it seems to disign us from what we like expect ourself to be doing right yeah there are several things that come to mind there one is that that we do have these nice um meta attentional models given by people like Lars and his 2021 paper which includes this deep parametric modeling so there is this kind of um what am I paying attention to how well am I paying attention and you can have this uh these deep generative hierarchies where you have attention over attention the other thing is I've been I've been this is a slightly separate Venture but I've been thinking about the uh the pragmatic value of attention and I think in some sense um there's a really nice paper by MIT and colleagues 2019 where they talk about um context dependent value driving attention so if I'm looking let's say for uh I mean I was talking about yesterday with juli and

kistin on the podcast if I'm in a crowd and I'm looking for my friend and he's Ginger I'm going to up uh up the Precision waiting of like a template of gingeriness which means that anything that's kind of Ginger whether it's a cat or my friend is going to go up and everything else is going to go relatively down and this comes back to original theories of um attention bi's competition Theory there more than Duncan but the idea is that that even my very attentional schema there is being governed by a high order pragmatic um the pragmatic gain of finding my friend and that can be there could be a lower order element of that which is that there is value in paying attention to these things because it it subserves that higher order uh inference cool all right I'll go to some questions in the chat but anyone's welcome to add more okay Steven CET asks my question is would you agree that self-reflective self self-awareness is necessary for time perception therefore time requires the self okay so this is this is an interesting question um I think in part it can be answered somewhat by the flow paper um so in the flow paper it really does focus on self- modeling and the idea here is that uh deep temporal planning in or involves the levels of the generative hierarchy which are propositional and cephal and those are the very levels are associated with a rich epistemic agent model so this um self-conception of uh an agent that can use its knowledge to uh from the past to inform uh future outcomes and it's it's future action policies now in flow because you have these shallow um you have these shallow decision trees you actually attenuate that part of the hierarchy now that means that when you're in flow people talk about the kind of self action mer we're not saying that the self is completely dissolved in flow States because for example um the minimal phenomenal self researchers such as um annel Seth or Felix blankenberg or yob linowski have grounded that an inter receptive now it's it's it's I think it's an open question um there are some people who think that all experience involves a fundamental sense of mindness um so someone like Julian kin has written a paper on this there are others who will say that um inter receptive inference is really at the heart of the minimal phenomenal self and if that dissolves then the self model itself might dissolve now whether you get time out of that is an interesting question I would Point people in the direction of a conversation I had with Carl where basically we had this there's this idea that this these different uh temporal hierarchies are predicated on the actual dimensions of the mark of blanket so how nested the mark of blanket of an organism is because if you're very far away from your actuators you have to start making inferences about what your action is versus what the world's action is so I think there's a nice link there between dimensionality maybe not just physical size but dimensions in the terms of embedded markof blankets and time I don't feel like I'm necessarily qualified to say um how

these two are necessarily like fundamentally connected because I think there's still an open philosophical uh and physics related question about the reality of time itself awesome cool you say a little more about trust and distrust it's very spicy to call the paper distrusting the policy because it feels like the interior dynamics of our thought should be like what we can trust so it's kind of a little bit of a stick in the wheel but that's kind of fun so where is there trust and where is their distrust in what in that kind of a cognitive model sure I mean uh this is rift off of yacob's distrusting the present so I want to clarify that as a as a master student you got to try and make some waves so it was just trying to get people's heads uh poking up but there is also the the idea of distrust and I think you can that as in the form of precision waiting um to go back to what yako was talking about in terms of this um trust as understood by philosophers we're talking about subpersonal basian beliefs which include Precision waiting and it's just that your Precision waiting over your state hypothesis is going to be downweighted over time because there's a higher order belief that the world is volatile and I think that higher order belief is really useful to think about when we're talking about what trust is in this respect if you thought that the world was wasn't going to be volatile such that for example you're staring at a blank wall then you could very happily trust that your hypothesis about that blank wall is going to be fine in one second two second 3 second 4 second the idea is that the the trust comes from this kind of um perhaps habitual inference that in this context I am permitted longer spans of uh deeper trees in this context however are those deeper trees are precluded now in some cases that might be full active inference so there might be a higher order in a deep parametric sense a higher order um action policy to deploy shallow action policies or uh temporally broad action policies in the state of flow we argued that this was habitual but I think the idea here is that for trust to trust to trust or not trust in action policy is not that we're not saying don't trust action right action is the fundamental currency in active inference we're saying you might want to agents have to be aware of the degree to which their action is going to yield them the outcomes they want not just here and now but also for expected free energy also for the subsequent steps awesome yes that clears it up it's not like policy selection isn't trustworthy it's that there's a variable speed at which policies like G stale that that totally makes sense one one really um interesting and open topic is which phenomena either objective behavioral computational type phenomena or or experiential phenomena do we associate just coarsely more with the A matrix and load more onto the Sens making component or more onto the B Matrix more into the world trans I model or the consequences of action in the world so like were there any phenomena that you saw different author's approach it like sometimes

discussing that phenomena in the context of a matrix and sense making other times more like in this action oriented B Matrix way honestly I haven't um for I think I think because deep parametric I might be missing something but I think because deep parametric modeling is a relatively new uh aspect of active inference a lot of the literature on this stuff especially in accounting for phenomenology adopt quite a strict or pure predictive coding language so if you read uh this distrusting the present this yakob and colleagues paper from 2015 there's no mention of the difference between the a m uh you know the uh the D Matrix and the B mat Matrix right the state this the inferred State um prior to observation and the state transitions based on observations which are contingent on action policies rather it's just that the prior is informed by observations and they talk about precisions of observations where I would talk about precisions of likelihood distributions so I would I I would also I would they don't touch upon this but I think uh it's definitely worthwhile me pointing people in the POS uh direction of this bogatar and Jabara paper um it can be a touch confusing because I think one difficulty that I was having is it it's hard to it's hard to hold both objective time and subjective time in your head at the same time I'm okay with one and then put then go to the other um but combining the two is a kind of amazing feat and they've done a great job so I would say just answering your question in terms of where did I see people providing different emphasis I would say it's difficult because I haven't seen that many deep parametric models of uh of time per se um and so that's one paper that I would Point people to but I think it's definitely a it's definitely a really interesting area of work about where we really put all the weight whether it's perceptual or action but I just didn't think that perceptual pure perceptual was going to do it for the flow of time at least not posited by um yakob and his team cool well um a few things strike me from this kind of presentation first how open and how many ways there are to address time phenomena in active it feels more like a toolbox or a mechanic shop for temporal or dynamical systems rather than giving us a singular answer like right off the bat we have discreet and continuous time now people have made Arguments for continuous time also people have made Arguments for discreet time and suffice to say that we are using a formalism that straddles both or can flip between both so there's many ways to describe it and another example of like how temporal phenomena can be brought into play differently is let's just say we were looking at a 1,000 step time series we could model that as a thousand steps of one or steps of 10 or we could do a nested model so that even though it's uh one time step model maybe the slowest level is the whole thousand time step right so it kind of coarsens the long time in to kind of compressing it into a perceptual moment even if an abstracted perception like is it possible to

perceive the 2000s or is it possible to perceive you a decade I thought was a really important part of this was the fact that we need some kind of objective order on which we can like rest our rest our arguments so you see in the paper and in the talk I really emphasize the fact that I'm talking per objective time unit and I'm staking I know again that's controversial right but like we just take as a kind of Miracle of our theory that we have seconds and minutes and hours if we can rest what we're talking about in terms of flows on that kind of um standard that we can then apply to different theories then Things become at least explicable because without that you're absolutely right it just depends on the the the level of the hierarchy you're at or or how deep you've gone into these time steps um but if you can frame them within something that we can all agree on then it becomes relative to that and I think it becomes uh the science becomes a lot simpler cool and also while certainly is nice and simplifying to take the chronology or the timeline as an external given there's also agentially constructed and then we might be looking for something like a gamma oscillation like we're able to distinguish intervals of time seemingly on the order of hundreds of milliseconds but not fractions of milliseconds and there seem to be some physical reasons like why we couldn't differentiate two things that happened within 1 millisecond for example so it's kind of like that gives us a bottom up granularity or at least atomicity for the time process that a given agent undergoes and for a tree that might be slower like its duty cycle might be slower and yet we could still use the same scale independent formalism to describe it yeah I mean I I wanted to make very clear in this that uh I am not capable of touching upon every element because this is one of the richest areas of philosophy and physics and I'm not a physicist um and to be H I'm not particularly well versed necessarily in the philosophy of time um what I wanted to do in this was stay really faithful to the phenomenology so this is informed really primarily by edman hle um but I think you're absolutely right I think an ultimate theory of time if there ever could be such a is going to have to integrate or find a bridge between the so-call objective and the so-call subjective and I think that is a qu that's up for grabs I mean you're absolutely right you can have a model where it's it's it's all emergent from um the actual agentic uh being so it makes me think of sort of Donald Hoffman I mean Donald Hoffman would just say that this is a screen you know this is a this is an interface that's really useful in biological agents but the underlying reality has no time um and of course you have physicists you know this notion of block time or eternalism that everything is everything is now every the past present and future is all real um again I I this is this is why I think it's important at least for me and other researchers when tackling these huge questions is demarcating exactly what you're going to be

talking about and uh I don't have the skill or the Acumen or the time if you pardon the pun to do the the Stein and the the call relli sorry chaps I I would have written a shorter letter if I had different time quite still counter I'll ask a question from live chat Sor is asks can the agent perceive time as the amount of parallel actions taken between events of attention I'll give a first thought on that y with what Chris field shared in the physics as information processing we talked a lot about the measurement and the preparation cycle and so in active inference at each time step sometimes we're just talking about like a vertical slice and there's an observation coming in and a hidden State inference it's all described kind of synchronously in the moment whereas in the quantum framework there's a bit more of a two stroke phase to the measurement which is that it's prepared and then measured and so if the external oscillator let's just say again give give us seconds for now if it's going at 1 second and an agent were preparing and measuring 10 times a second well then nine of those 10 would kind of come back empty like the nine sequential measurements and proposal preparations and measurements that didn't have any change would be kind of like in a Timeless or same time moment they all would have like pinged the clock and gotten the same state back um and then conversely an agent could use the rates and the type of change or interaction with the environment as a proxy of change and the example I think of there is like honeybees using the optic flow as part of their distance calculations and people have shown like if you have the honeybee move through a tunnel with very rapid black and white alternation it overestimates the distance it flows right and then if you have it in large contiguous color blocks it underestimates the distance it flows so it's using the rates of optic flow change frame differencing which is also the origins of predicted processing using those kinds of external oscillations or changes is as part of its embodied time perception right yeah I the way I would approach this is exactly what you said which is when we when we build a base graph something that s of became very clear to me is that we we are taking time we are slicing time which is already which is a detriment because I'm trying to talk about dionic time whilst slicing my model and I think this really does just exhibit sometimes the Discord that we have between our modeling and the actual reality so it's it's an inherent weakness or an inherent limitation of our instruments what I will say is I think it's interesting to talk about uh parallel processing rather than serial processing here because as I keep trying to make clear this generative hierarchy is rather deep and so I think for example you know I mentioned when one stares at a wall there's always some State updating right the the game is never the the game has never paused and so I think there's actually an interesting follow-up question here which is if

we stack all of these regimes of attention up and we have rapid flows in all of them right does that lead to an even more pronounced sense that time is flowing faster in the way that maybe now we can have this nice hierarchy of what I could predict uh stable predict uh stably predictive d uh situations dynamically predictive situ and flow where the difference really is the actual number of attentional regimes in which you're seeing updates because I don't there is bear in mind that to update the C Matrix you have to know where you are in the first place you have to be able to do the state uh updating it's just that for me that was insufficient to explain the phenomena at hand but if you can now start taking that kind of base graph that I showed uh where is it you know if we can take this Bay graph and in principle apply it up and down the generative hierarchy well if these narrow um flows are are happening all the way up and down maybe there's an interesting thing there that this is really you know the the the top end of Passage phenomenology and rather and if these are really protracted at every level right right down to the very lowest markof blanket of our system and right up to the move rather slowly so I think there's some interesting ideas there um and I think it it feeds into this notion that we we occupy or embody embedded markof blankets that go beyond us and in many ways go kind of beneath us I mean it depends on where you cast your identity but yeah I think the parallel thing here is is really interesting but for instrumental scientific reasons it was just a lot easier for me to like and philosophical reasons just to take one slice at one level of the hierarchy although this does have some parametrization cool okay so looking at the figure from your paper yeah so this one but but by the way this one's from the first flow paper okay I think this one yeah so this one's from the first flow paper um I'm not sure I need to check I'm not sure it makes an appearance in the second one but it's it's it's that's where it comes from okay so for any parameter that you know or love on this this diagram yeah what what would you say is like a prompt or a cue that might help us understand or come even to an intuitive felt sense sure pick pick anyone because I think that's like super interesting and it's all awesome to look at this as like the compilation instructions for a base graph which is in one sense what it is technically however also as we're talking about our own minds and experiences here it's like super fun and definitely of interest to tap into whichever letter and then let's see if we can kind of go from there yeah so as a sort of just forewarning um this was like yeah as I said constructed with Lars I mean L's really LED on this project I should give him a lot of credit um and so I also don't want to butcher his work because I know he's really into this sort of deep parametric modeling that he learned from KL but you it's my paper so I have a good idea what's going on here um I think I could do an okay job of actually explaining

what's going on more generally um which is the idea here is that let's just start right at the top at the top is not particularly it shouldn't be too confusing which is that basically you're just inferring the state that you're in right so what do you mean by state so the hidden state right so I'm inferring um so this um given this observation of you know whatever has hit your receptors I believe with some high Precision that I in a uh you know I'm about to start a violin concerto okay let's go with the violin violin kerto situation and just so I'm about to start playing my violin now what does that mean well that feeds into this e Matrix um this e with the two as the superscript and so what we're saying here is is that that very State inference acts as a contextual cue for the deployment of this action you uh with the superscript two and subscript one so really this is all basically driven by this e Matrix e with the two superscript now what does that look like well what's really important here is this s with the superscript two and subscript two because what they what we're saying here is is that we we're we're we're posing a mental action and the mental action is to increase per decision over the likelihood mapping the trans so the likelihood mapping is here the transition mapping and the uh position of preferences so what we're saying here is that this action is basically it in some sense it this is its kind of preferred State um now the only issue as I said is we can't observe Precision what we can observe are the outcomes of our action so what we're now saying here is that basically this observation here through this likelihood Matrix will feed back up the hierarchy and inform the mental state inference here oh you've you've yes you've succeeded in increasing precision and the reason for that is as you know if we get increased Precision over the C Matrix and the B Matrix we are driven by pragmatic action at least according to the expected free energy formula so that's kind of what's going on here um note to is there anything else that's kind of relevant um yeah I mean you so you got to bear in mind that like so again this is uh mental action but these are the these are the physical actions so these are the actual playing of the violin right and they're going to contribute to this observation but this obs is really in terms of mental state inference doing the job of saying you've done a good job at increasing Precision over your sensory expectations the likelihood Matrix and transition um mappings let let me um try to try to Echo this yeah we're in the violin setting so at the purple top layer there's two contexts there's the presentation in the auditorium that's the real thing and then there's the practice at home so there's two contextual States and and 90% of the time you're at home so that's your prior but then sensory evidence suggests that you're in the auditorium so that's the context um the blue layer is the action policies like the angle and the motor um activity of the arm and the Hand of the violinist this is this is yes this is the physical actions yeah yeah and

the Orange is the regime of attention of the musician now right what they expect in terms of recital from sheet music yeah is that pragmatic value is playing like the most amazing and Transcendent version of the sheet music as written so of course the map is not the territory however pragmatic value is defined as hitting those notes at the right order in the right time so it's not like an epistemic exploratory um place so now um how how do we think about how those motor movements are skillfully done which can include not just deafness but also planning like um a wind player has to plan their breathing and there's other things and planning the um motions all kinds of things and so um your kind of contention with the flow state is that in a skilled um cant mahol setting y where there's Precision like yeah I am good at skiing I am good at violin personally not but I'm just saying um yeah yeah yeah when confirmatory evidence is coming in then that enables a shorter time Horizon of planning so A vastly reduced cognitive overhead but relying on tacit and implicit heris discs that enable um professional outcomes to arise in that flow State even if somebody looking from the outside thought oh wow they must have really planned that yes however it wasn't really planned and then we kind of have the orange level the regime of attention meeting in the middle and mediating this like cultural contextual understanding top down with the body bottom up um sensory component yeah so that's that's that's basically it um there there's plenty of things to point out there one is um I had this idea that these observations that you're making sort of in real time are actually basically Loop Round into this contextual State inference so at time n plus one it actually kind of feeds it into a conversation about time what I've actually just observed so the very sound that has come out of my violin itself is feeding the contextual State inference which itself feeds another habitual action to cast Precision over sensory expectations about the next thing that I'm going to play and then that Cycles so we're basically saying that although the actual um action policy the actual selection of the actual policy here at C1 is pure active inference the mental action that drives High Precision over it is Habitual and that itself is driven in this recursive way so that's one thing yeah the other thing is is about this kind of shallow temporal planning that's exactly right that's exactly what we propos the reason for this and this was actually raised by yakob har that's I'm very grateful for his observation here and this is actually mentioned in his um 2022 self- evidencing paper is that um the like it's not the case that when I play my violin skillfully I can say okay I know I have it's not the case I believe that everything I do is going to be highly predictable for 3 or four minutes cuz that would afford deep temporal planning rather as I said there are these volatilities in the environment that I always need to be adjusting myself to right I need to be conforming myself

to I need to be able to exploit those affordances so flow states are not just one extended action policy which is super rigid rather it's it's kind of because because each state uh inference ex itself acts as a contextual cue over certain sensory expectations we have this kind of uh fractal nature where you might have like the first state well might be fractal but it's like a tree you have the first a inference and then the and then you have these bations such that sure everything is um constrained within the piece that you're actually playing but it's not going to be the same piece every single time now what we do say is is that if I have an observation for example if my string suddenly breaks what happens is this with this Bas graph is it it basically collapses and and the reason for that is is that that lower level perceptual State inference which feeds into the context I'm in is no longer a sufficient context to trigger the habitual mental action so it's not just the case that you can be in low in any context right that's really important um there's a set of uh contexts which are going to license this mental action and that's very nice to explain within active inference as just hashing uh action policies as habits so the more you do them uh the more they just get cached and the sort of um more precise that is as a KNE Matrix so that also allows us to talk about cognitive effort in line with Thomas PA's 2023 paper about sort of it being divers uh you know effort being Divergence from PRI uh prior over policies but yeah I mean the fundamental Point here is that flow is a weird one because everything is like very predictable within its ra within its action policy regime but that's it the skill really is being able to have like a diverse or fractal uh structure where you can go to different nodes and they all link together through habits basically awesome okay few fun questions in the chat about Precision get your Precision goggles on I hope you're confident yeah okay Dave writes yeah quote we can't observe our precision he says this assertion seems to need clarification or qualification eventually our precisions have profound effects on our actions and anxieties they must need attention somehow and in fact affective active inference have posited that there's something like the observation of a variance parameter and that that's associated with anxiety and veilance by hesp at all so which variances do we imagine are introspective and why so that's a good question um at least in the way that this I I don't think to be honest it makes a dramatic difference whether you have another level within this base graph which has the observation of precision our idea here is that just because things are happening very quickly it it might just be parsimonious to invoke that the evidence for that Precision waiting comes from the observations so that's that kind of oh um with a two as a subscript and one as a superscript um but no I mean I I'm not going to I'm not going to die on this on this hill uh because also I don't think it makes a massive difference um I might I would

be personally inclined if he was here to pass this one off to lar um because that was the way he explained it to me and I'm not trying to exculpate myself but recall that this paper is a six or five auor collaboration so it's not all me but that's the way that I would say I would say that there's no harm in saying that you can observe your preferences oh sorry you're Precision waiting but in some sense we're treating this extra receptively um and so yeah this this would this seems to work um in a way that doesn't need to invoke a another layer to the hierarchy great that's not a hill to die on that's our research field yeah exactly but but seriously it's a great question by Dave and those are the questions that we could imagine real studies to ask or address yes so it's it's those are the great questions okay yeah I mean the other thing I would say here is something that's become quite obvious to me again I mean I'm not the expert on this you might want to ask Thomas par but something that's become very obvious to me is that there's a difference we got to make a difference between attention precision and precision waiting um and I think there's often times in the community a conflation of those three and so when we saying okay we need to pay attention to these certain things well atten in the a matrix um which is a certain type of precision but we have other precisions in mental state in mental inference including these C matrices B matrices e matrices and d matrices and they do not all map onto attention so this idea that Precision waiting is attention which is kind of an okay heuristic it's not really what's going on yes the terms not not to make like too general of a statement here but the terms that refer more generally to statistic like State belief variance attention Precision confidence usually without more qualification they're too vague like just saying something as a prior are we talking about D in the purple D2 D1 are we talking about B playing the role as the prior for a future time step so um like they are course grings their fingers pointing at moons and then there's the movement of the solar system and there's a lot of pointing and gesturing and also thankfully rapid accelerations of our understanding and our tool environment to help like move this model for example into an executable format or to use verbal descriptions of it and there's some really exciting and fun things like even just pasting this image or uploading the paper into a language model can give some fun and different insights now it may take an expert to then evaluate whether what it says is accurate or not however that's a super promising Avenue for like developing context and metaphors or examples for looking at this yeah I mean I need to add that like I have I I really do not I'm not treating this as like a way to have a go at philosophers because my background is in philosophy and cognitive science so this emphasis on the computational model in many ways actually owes a lot to Carl because when I presented this flow paper for the first time in his

group he was like well it's sounds sort of philosophically narration good but kind of the emphasis now is on like can you put this into M lab can you build something from it um that said I mean as I said this is informed fundament this paper at least was informed fundamentally by lived experience and phenomenology so I guess what I'm trying to become a part of is the so-called um generative passage uh which is finding a way to get from phenomenology to basian mechanics and hopefully this is one way but I don't I don't I don't I don't propose that I'm reducing the rich phenomenology of lived experience to these basium mechanics um because I think that would be slightly humoristic because no one knows yeah it's not even just approaching it from both sides I don't think there's only two sides to this yeah there is though this just like you said grounding in the felt experience and the phenomenology the the ontology and the practices of philosophy and also formalism and wanting to touch grass but touch Matt lab as it were yeah and then that kind of strange looping when our felt experience H has a reference of a cognitive map and model right and that kind of special a yeah yeah there's a tripartite structure here which is one the basium mechanics two the uh the model and three the phenomenon the target phenomenon and I think again this is actually really uh become apparent to me after speaking to Julian yesterday Julian kin that one has to be careful to not fall into the tendency to conflate them or reduce one to the other as you say uh the map is not the territory is a really crucial component of all of this I mean I can provide a generative model I can speak about C matrices I I I I really want to clarify do not mean that in a reductive sense that the flow of time is a c you is the updating of the C Matrix what the actual binding the relation is there I don't know um whether it's emergent strongly emergent weakly emergent epiphenomenal is ontologically reductive it is is is a good question um but it's an open question I don't think we should just take for granted that they are one and the same thing that we're making an identity claimer yeah of course many interpretations and we welcome them I see this as like a a boundary artifact or a design artifact that brings us to the point of ju's deposition of all of those schools of thought that you just mentioned in more and then it's like through the bottleneck together maybe the distribution of beliefs on the other side is the same maybe it's different right but we came together we synchronized our generative models and then we get to continue the discussion like from a point of contact or alignment otherwise different assertions might be made that don't even have the same underlying axiomatic set or refer or inclusion Criterion at which point you might as well have one person saying well I designed a pickup truck and someone else is arguing about whether my red is your red and someone else talking about ecology it's like it would be like blending

up pages of a library and not it's very yeah it's very important to to keep a philosophical eye on on the theories that are emerging and and to be honest Daniel I mean it's credit to you I think as a younger person entering into active inference you guys have all done a very good job at not making us subscribe to one philosophical position um one ontological EP epistemological position I mean people have their own I for one I'm personally happy to leave it open I mean I say that before doing my PhD I'm sure after my PhD I'll be an acolyte of some of some way of thinking but I think this is the exciting thing about active inference and hopefully that enthusiasm comes across in the podcast which is like it kind of works whether you're a reductionist and an epiphenomenalist like it's just maths and statistics and it's it's informative for philosophical conversations and so philosophical conversations are informative to the mass and statistics and framing it but it doesn't it doesn't constrain you like um maybe having a super radical and activist cognition account might constrain you or being a pure computation list might constrain you so so uh credit to everyone credit to credit to car credit to you no it's it's good thank you that thank you it's awesome and it makes me wonder there's so many ways to go from there but how much of this is active inference itself if anything could even be itself and how much is merely optimal grasp of formalism and just seeing that whatever you write on the floor the table or the page any equation category Theory dynamical systems statistics like any anything is the diving board that you have to jump into the pool with for phenomenology and so what's the point of comparing diving boards only if it's about dive you'll notice I didn't even mention active inference I mean like I mean one because I'm we're here with the Institute it would be a bit silly to waste time explaining what the free energy principle is um I mean from my perspective the way that I talk about it is well I'll take the free energy principle is just unfalsifiable true as a principle of physics it's like it doesn't really get more fundamental than the free energy principle where it really gets interesting is the fact that if I am to be considered the same agent or same organism from time one to time two then I need to be abiding by the free energy principle now the real question is how am I doing that and predictive processing or PP seems give a really nice weigh in but I kind of just take it that like I'm doing some form of active inference all the time and like from that I let it take me where it takes me whether I stick purely to a philosophical account or whether I stick to a more computational account like none of my commitments there are negating my ultimate commitment to my uh you know the the everpresent reduction of variational free energy so it's a nice building block but as you say I think uh the lovely thing about it is it can lead you in different Direction sure it might be I don't want to say that it's just

an Open Season people have their commitments so if you're a predictive coding guy you have your commitments if you're a variational message passing guy you have a commitments for me I'm not I hav I really don't have any skin in the game at that level because I've been in this game for a year so um I'm happy to I'm happy to see where things go and and and just yeah see how and for me it's really important that it maps on in some way to the phenomenology if you're talking about something that clearly like just from Pure analysis of my own experience is not the case well then then that's problematic interesting again I I love this topic and there's many things we can continue on and it makes me think about how coming from an animal behavior and genetics background where firsters experience was either implicitly or explicit all of the phenomenology feels special to bring in kind of like the leavening of the bread a little bit because the animal behavior perspective is like everything but this right even the orange layer here would be kind of like a a proposal for a cognitive analysis but only to get a better handle on the blue never because there was something intrinsically like meaningful or experiential it would be just too speculative and a field so it's just interesting to come from the orange layer and descend into the blue and get into the nitty-gritty with the motor reflexes and the iades and the way that philosophers get into these embodied phenomena and then also from the embodied in the behavioral scientist side to come up into be like yeah actually we can't s side step some of these philosophical questions it does matter how these different systems perceive and engage with time yeah I mean I think it's important to bear in mind that um the behaviorists for example never they never they never actually excluded in terms of ontology uh in their case representational States they just said that like they're uninteresting to science because they're unobservable so like an animal someone studying ANS as you do or bees they're behaviorists but that actually doesn't that doesn't um that doesn't commit them to saying anything about the representational capacities of a be or an ant um but yes I mean these philosophical questions are interesting I'm personally happy to start at the insights that people like Merlo Ponte or hle or someone like Julian have derived from these very precise I mean phenomenology is not just like a random science it's not woohoo like it's it's it's codified categorical and I'm happy to start with that and then when I find something like uh the that clearly violates what at least seems to be pretty obvious in my experience now to go to your experience would require some philosophically slightly uh Troublesome inferences because you know you could be a philosophical zombie and I'm not zombies um but at least for my experience you know when people you know people get very angry about uh Illusionist about Consciousness not realizing that they're not actually Illusionist about Consciousness they're

illus Illusionist about phenomenal States or phenomenal qualities but if you for example you were an Illusionist about Consciousness wholesale well I just I I can't start there with you because you're saying something that is just so clearly wrong so again I'm not going for Daniel denner or Keith franker because I know they actually don't say that but for me it's a decent starting point to say well I have very intimate uh knowledge of that which is happening subjectively and let me see how close I can get there with computational schema whilst not losing the phenomenology I think that's the important thing and then I think where you end up emphasizing is just like what your interests are right like I'm fundamentally a philosophy guy I'm not a computationalist by training as you can see probably from my explanation of this graph um so I've been really lucky right I've been really lucky to work with people like lar and Carl who have kind of helped me and I think it's really really useful but again I'm not going to lose sight of lived experience um and I don't think anyone personally I don't think anyone should but it's difficult as as you say when you're studying animal behavior because well that's that seems like a stronger inference to make because maybe they don't have public language to say what it's like to be an ant or a beast to paraphrase Thomas nle well very fun um what are your next steps or directions so yeah so this is up on the preprint as I said I want feedback I want people to get angry and attack me uh or be it in the spirit of scientific consilience um so I'm I very much encourage feedback I think um I'm hoping to get some more formal reviews from some people and then we'll try and get this out there somewhere I don't know quite where I'm working on some other things um which I'm sure I can speak about I'm working on a paper about self- modeling in Tourette with liby s which is super exciting um so that's coming together I'm working I'm at the very early stages of working on something about extended mind uh with Ma and others so that should be really fun and the podcast um we've been putting in loads of work on the podcast so we have Chris rff which is super exciting just a sort of Pioneer in Psychology uh Julian kin as I mentioned Alex or Robia in a wonderful conversation about biometrics and cybernetics and I learned loads and Lance the Costa and that was the one where I was telling you before we went on live there was a lot of maths learning so to be honest Daniel I'm just learning um and I'm applying for phds and hopefully that will kick off either later this year or or the new year but I'm not going anyway I'll put it that way awesome very fun yeah very epistemic very chill so very chill yeah a lot of good comments in the live chat you can review so thank you um everyone in the live chat thank you Darius and thank you everyone see you all thank you thank bye